

Assignment 1

Wireshark Packet Capture & Protocol Analysis

Objective

To capture live network packets using Wireshark and identify basic network protocols such as TCP, UDP, IP, and DNS.

Procedure

Step 1: Launch Wireshark

- Open Wireshark
- You will see a list of network interfaces:
 - Wi-Fi
 - Ethernet

Step 2: Start Packet Capture

- Double-click on Wi-Fi (or active interface)
- Packet capture will start immediately

Step 3: Generate Network Traffic

Perform any of the following:

- Open a web browser
- Visit any website (e.g., google.com)
- Open Command Prompt and run:
 - ping google.com

Step 4: Stop Capture

- Click the red Stop button
- Captured packets will remain on the screen

Step 5: Apply Basic Filters

Use the Display Filter bar:

Filter	Purpose
ip	Show IP packets
tcp	Show TCP packets
udp	Show UDP packets
dns	Show DNS packets
icmp	Show ping packets

Step 6: Analyze a Packet

- Click on any packet

- Observe:

Header	Observed Value
Ethernet Header	Ethernet II, Src: Intel_9c:22:64 (84:5c:f3:9c:22:64), Dst: d6:2f:ca:ea:30:64 (d6:2f:ca:ea:30:64)
IP Header	Internet Protocol Version 6, Src: 2409:40d1:1e:14b1:e5f9:1f61:c36d:157d, Dst: 64:ff9b::36c0:8e02
TCP/UDP Header	Transmission Control Protocol, Src Port: 50707, Dst Port: 443, Seq: 1, Ack: 1, Len: 1
Payload Data	Application layer data (HTTP payload)

Observation

Parameter	Value
Interface used	Wi-Fi
Protocols observed	TCP
Source IP address	2409:40d1:1e:14b1:e5f9:1f61:c36d:157d
Destination IP address	64:ff9b::36c0:8e02
Packet length	21 bytes

Result

Live network packets were successfully captured using Wireshark, and basic network protocols were identified and analysed.

Apply a display filter ... <Ctrl-/>					
No.	Time	Source	Destination	Protocol	Length Info
229	12.222225700	64:ff9b::34b8:dee4	2409:40d1:1e:14b1:e...	TCP	510 443 → 59684 [PSH, ACK] Seq=3661 Ack=2042 Win=64128 Len=436 [TCP PDU reassembled in 232]
230	12.222339200	2409:40d1:1e:14b1:e...	64:ff9b::34b8:dee4	TCP	74 59684 → 443 [ACK] Seq=2042 Ack=4097 Win=65280 Len=0
231	12.223271800	64:ff9b::34b8:dee4	2409:40d1:1e:14b1:e...	TCP	2514 443 → 59684 [PSH, ACK] Seq=4097 Ack=2042 Win=64128 Len=2440 [TCP PDU reassembled in 232]
232	12.223271800	64:ff9b::34b8:dee4	2409:40d1:1e:14b1:e...	TLSv1.3	1128 Application Data, Application Data
233	12.223366600	2409:40d1:1e:14b1:e...	64:ff9b::34b8:dee4	TCP	74 59684 → 443 [ACK] Seq=2042 Ack=7591 Win=65280 Len=0
234	12.223663800	2409:40d1:1e:14b1:e...	64:ff9b::34b8:dee4	TLSv1.3	154 Change Cipher Spec, Application Data
235	12.424901100	64:ff9b::36d0:e900	2409:40d1:1e:14b1:e...	TCP	86 443 → 49787 [ACK] Seq=1 Ack=2 Win=205 Len=0 SLE=1 SRE=2
236	12.424901100	64:ff9b::34b8:dee4	2409:40d1:1e:14b1:e...	TLSv1.3	680 Application Data, Application Data
237	12.425023400	2409:40d1:1e:14b1:e...	64:ff9b::34b8:dee4	TCP	74 64791 → 443 [ACK] Seq=3113 Ack=8197 Win=64768 Len=0
238	12.430919000	64:ff9b::34b8:dee4	2409:40d1:1e:14b1:e...	TLSv1.3	371 Application Data
239	12.480871900	2409:40d1:1e:14b1:e...	64:ff9b::34b8:dee4	TCP	74 64791 → 443 [ACK] Seq=3113 Ack=8494 Win=64512 Len=0
240	12.491376300	64:ff9b::34b8:dee4	2409:40d1:1e:14b1:e...	TLSv1.3	680 Application Data, Application Data
241	12.491473600	2409:40d1:1e:14b1:e...	64:ff9b::34b8:dee4	TCP	74 59684 → 443 [ACK] Seq=2122 Ack=8197 Win=64768 Len=0
242	13.113333000	64:ff9b::9765:4334	2409:40d1:1e:14b1:e...	TLSv1.2	893 [TCP Previous segment not captured] , Ignored Unknown Record
243	13.113392200	2409:40d1:1e:14b1:e...	64:ff9b::9765:4334	TCP	86 [TCP Dup ACK 140#1] 64546 → 443 [ACK] Seq=32 Ack=16159 Win=255 Len=0 SLE=17379 SRE=18198
244	13.121380700	64:ff9b::9765:4334	2409:40d1:1e:14b1:e...	TCP	1294 [TCP Retransmission] 443 → 64546 [ACK] Seq=16159 Ack=32 Win=421 Len=1220
245	13.121457900	2409:40d1:1e:14b1:e...	64:ff9b::9765:4334	TCP	74 64546 → 443 [ACK] Seq=32 Ack=18198 Win=255 Len=0
246	13.417235500	172.20.10.4	224.77.77.77	UDP	155 50572 → 12177 Len=113
247	14.359884200	64:ff9b::9765:4334	2409:40d1:1e:14b1:e...	TCP	1294 443 → 64546 [ACK] Seq=18198 Ack=32 Win=421 Len=1220

Frame 1: Packet, 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface \Device\NPF_{DFB4267F-60B7-40F4-888D-1A766255}

Ethernet II, Src: Intel_9c:22:64 (84:5c:f3:9c:22:64), Dst: d6:2f:ca:ea:30:64 (d6:2f:ca:ea:30:64)

Internet Protocol Version 6, Src: 2409:40d1:1e:14b1:e5f9:1f61:c36d:157d, Dst: 64:ff9b::36c0:8e02

Transmission Control Protocol, Src Port: 50707, Dst Port: 443, Seq: 1, Ack: 1, Len: 1

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C:\Users\susha>ipconfig
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Windows IP Configuration
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Ethernet adapter Ethernet:
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Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
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Wireless LAN adapter Local Area Connection* 1:
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Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
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Wireless LAN adapter Local Area Connection* 11:
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Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
```

```
Wireless LAN adapter Wi-Fi:
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Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2409:40d1:1e:14b1:d9a3:1c8:f5b9:7c92
Temporary IPv6 Address. . . . . : 2409:40d1:1e:14b1:e5f9:1f61:c36d:157d
Link-local IPv6 Address . . . . . : fe80::aa3c:15d9:19f2:f565%20
IPv4 Address. . . . . : 172.20.10.4
Subnet Mask . . . . . : 255.255.255.240
Default Gateway . . . . . : fe80::d42f:caff:feea:3064%20
```

```
Media State . . . . . : Media disconnected
```

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Pinging 172.20.10.4 with 32 bytes of data:
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Reply from 172.20.10.4: bytes=32 time<1ms TTL=128
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Reply from 172.20.10.4: bytes=32 time<1ms TTL=128
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Reply from 172.20.10.4: bytes=32 time<1ms TTL=128
```

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Reply from 172.20.10.4: bytes=32 time<1ms TTL=128
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Ping statistics for 172.20.10.4:
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Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
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Minimum = 0ms, Maximum = 0ms, Average = 0ms
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